National University of Computer and Emerging Sciences
Lahore Campus

Introduction to the Internet of Things (IO3031)

Sessional-II Exam

Date: 9th April, 2026

Course Instructor

Ms. Abeer Bashir

Total Time (Hrs):	**1**
Total Marks:	**30**
Total Questions:	**3**

Roll No ____________ Section ____________ Student Signature ____________

Do not write below this line

Instructions:

1. **Do not forget to write your Name and Roll Numbers.**
2. **Make assumptions if you think some data is missing.**

CLO # 2: **Apply the knowledge of Internet Connectivity Principles, IoT Networking Technologies, Protocols, and standards (PLO 1,C3)**

Q1:

a. An IoT system is developed using an ESP32 or ESP8266 , where the device connects to a cloud platform (e.g., Blynk, Firebase) over Wi-Fi as performed in the lab. The device periodically transmits the string "Internet of Things" to the server, where it is stored. **Determine** and **list** the protocols used at each layer of the communication system between the IoT device and the cloud server.

[5+5 marks]

REST

Layer	IoT Device	Server
Application Layer		
Transport Layer		
Netwrok Layer		
Link Layer		

HTTP

b. **Examine** the differences between traditional networks and software-defined networks (SDN). **Illustrate** your answer with appropriate labeled diagrams and **list** three key differences.

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CLO # 2: Apply the knowledge of Internet Connectivity Principles, IoT Network Technologies, Protocols, and standards (PLO 1, C3)

Q2: An IoT-based smart parking system is deployed in a large parking area, where the area is divided into multiple slots. Each slot is equipped with sensors to monitor its status (empty or occupied). The sensors periodically detect the status and transmit this information reliably to a cloud server for monitoring. The cloud service provider offers three communication protocols: HTTP, MQTT, and CoAP, on a paid basis. The cost of using these protocols is as follows: [10 marks]

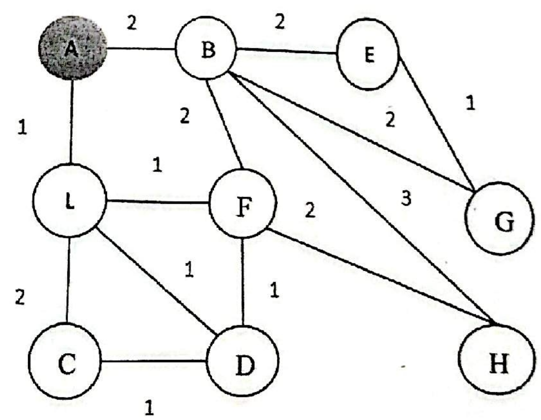
- HTTP (highest cost)
- MQTT (moderate cost)
- CoAP (lowest cost)

Considering the large parking area with many slots, an efficient system is required that ensures relatively low cost, minimal energy consumption, and reliable end-to-end data communication. **Determine** and **select** the most suitable protocol for this system. Justify your selection with three valid reasons.

CLO # **2: Apply the knowledge of Internet Connectivity Principles, IoT Network Technologies, Protocols, and standards (PLO 1, C3)**

Q3: For the given Low-power and Lossy network (LLN), construct DODAG. The Objective Function is as follows: Metric: Link cost, Rank=Hop count. **Construct** the complete DODAG, providing detailed calculations and explanations for each step until the final structure is established.

[10 marks]



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National University of Computer and Emerging Sciences
Lahore Campus

Introduction to the Internet of Things (I03031)

Sessional-I Exam

Date: 21st February, 2026

Course Instructor (s)

Ms. Abeer Bashir

Total Time (Hrs): 1
Total Marks: 30
Total Questions: 2

Roll No Section Student Signature

Do not write below this line

Instructions:
1. Do not forget to write your Name and Roll Numbers.
2. Make assumptions if you think some data is missing.

CLO #1: **Use understanding of IoT architecture and identify practical application areas (PLO 1, C3)**

Q1: From the system details given below, perform the following tasks: [15 marks]

1. **Examine** the system description, list the following components and explain their roles:
   i. IoT devices/endpoints
   ii. Controller services
   iii. Observer nodes
   iv. Cloud analytics
2. **Determine** the IoT Level of this system and show proper justification.
3. **Draw** the labeled diagram clearly illustrating the components and data flow of the selected level for this system.
4. **Classify** the system functions into:
   i. Local processing
   ii. Cloud processing
5. **List** the APIs that can be used to send the sensor data to the cloud platform.
6. **Examine** the difference between the APIs you have listed in part 5 with the help of diagrams.

System Detail: A smart irrigation system is deployed on a farm. The system includes:

a. Soil moisture sensors installed in different zones.
b. Each monitoring node consists of :

i. Soil moisture sensor

ii. Microcontroller
iii. Relay-controlled water pump
iv. Communication module

c. Each node performs local threshold-based control of irrigation and visualization.
d. Sensor data from each node is sent directly to a cloud platform where the data is stored, visualized and analyzed.
e. A mobile application accesses the cloud platform to monitor trends and remotely control the system.

CLO # 1: Use understanding of IoT architecture and Identify practical application areas (PLO 1, C3)

Q2: Write a python code to simulate an IoT system that collects temperature data from multiple sensors, send alerts if temperature exceeds a threshold and stores exceeded temperatures in a separate list for record. (Threshold value is 40°C) [15 marks]

The code should do the following tasks:

a. **Construct** a dictionary which store readings from 5 sensors.
b. **Acquire** the temperature readings for each sensor from user.
c. **Write** a function that checks the temperature of each sensor. If the temperature exceeds the threshold, prints an alert message and stores that sensor reading in a list for record.
d. Count and **show** the number of sensor readings exceeded the threshold and the number of readings that are normal.
e. **Finally, calculate** and **show** the maximum, minimum and average temperature of the sensors in the list.

NOTE: Clearly write your code with comments.